

ERAY KULKIZAGA

Software Engineer | AI/ML & Computer Vision | Backend AI Systems

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Software Engineering graduate specializing in real-time AI, computer vision, and ML deployment. Experienced in training detection and recognition models, building landmark extraction and ONNX inference pipelines, and serving low-latency AI systems through FastAPI and WebSocket-based architectures.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, ABAP

AI/ML & Computer Vision: TensorFlow/Keras, OpenCV, ONNX Runtime, YOLO-OBB, MediaPipe, RTMPose/RTMW, Landmark Extraction, Pose Estimation

Backend & Real-Time Systems: FastAPI, REST APIs, WebSocket, Pydantic, SQLAlchemy, Low-Latency Inference Pipelines

Data, Tools & Deployment: PostgreSQL, MLflow, Docker, Git, GitHub, Google Colab

Frontend & Visualization: React, Three.js, React Three Fiber

EDUCATION

Eastern Mediterranean University — Famagusta, Northern Cyprus

B.Sc. in Software Engineering

Sep 2021 – Jun 2026

- ABET-accredited | 100% English-taught Program
- Dean's Honor List and Dean's High Honor List recipient | Final Semester GPA: 3.83/4.00
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Data Science, Linear Algebra, Probability, Discrete Mathematics, Machine Learning, Embedded Systems

SELECTED PROJECTS

Sentinel-X | AI-Powered UAV & Remote Sensing Platform | [GitHub](#)

Jun 2026 – Present

Tech: Python, FastAPI, Pydantic, WebSocket, React, TypeScript, Three.js, YOLO-OBB, ONNX Runtime, OpenCV, Docker

- Built a full-stack UAV mission-awareness platform for search-and-rescue workflows, combining real-time telemetry, sector coverage, candidate tracking, uncertainty handling, and 3D mission visualization.
- Trained and exported a YOLO-OBB model on DOTA remote-sensing data, mapping labels into 5 production classes such as aircraft, vessel, and vehicle.
- Implemented an ONNX Runtime + OpenCV inference pipeline for frame preprocessing, object detection, postprocessing, latency tracking, and structured detection evidence.
- Developed FastAPI REST/WebSocket services with mission snapshots, model loading, evaluation routes, MLflow-tracked metadata, and a Pydantic-validated LLM Mission Copilot.
- Achieved 95.01% mAP50, 79.23% mAP50-95, 93.24% precision, and 88.92% recall in YOLO-OBB evaluation on DOTA-based remote-sensing data.

SignaTurk | Real-Time Turkish Sign Language Recognition System | [GitHub](#)

Nov 2025 – Jun 2026

Tech: Python, TensorFlow/Keras, RTMPose/RTMW, ONNX Runtime, OpenCV, FastAPI, WebSocket, NumPy

- Trained a 226-class Turkish Sign Language recognition model on the AUTSL dataset, processing 35K+ videos through a custom video-to-landmark and sequence-classification pipeline.
- Achieved 94.17% Top-1 accuracy, 99.49% Top-5 accuracy and 94.00% macro-F1 in the final pipeline.
- Built the landmark extraction pipeline with RTMPose/RTMW and ONNX Runtime, converting RGB video frames into structured body and hand landmark sequences for model training and inference.
- Implemented the FastAPI + WebSocket real-time inference backend, handling live camera frame ingestion, preprocessing, sequence buffering, model execution, confidence scoring, and timing diagnostics.
- This project also earned the award of "Outstanding Project" among the 30 graduation projects of EMU.

PROFESSIONAL EXPERIENCE

Microsoft | AI Innovators Summer Intern

Jun 2026 – Aug 2026 (Ongoing)

- Developing quantum computing fundamentals through Q# and the Microsoft Quantum Development Kit, including gate-level circuit design and quantum state preparation workflows.
- Applying computational logic to translate classical problem-solving approaches into quantum-ready algorithmic structures with a focus on modularity and scalability.
- Managing project work through GitHub, technical documentation, and collaborative development practices while preparing research-based technical outputs.

FlyRank.ai | AI Intern

Jul 2026 – Sep 2026 (Ongoing)

- Participating in a remote AI internship focused on AI-powered search visibility, SEO/AEO workflows, content optimization, and organic growth strategies.
- Using modern AI tools to support research, automation, content workflows, and structured problem solving across business-oriented use cases.

Iskur Holding | SAP/ABAP Software Engineering Intern

Jun 2024 – Aug 2024

- Engineered optimized SAP ABAP data-fetching workflows and Open SQL JOIN queries for enterprise datasets, reducing data processing time by 41% from 120s to 70s.
- Developed Contract Tracking and Barcode Screen modules with automated verification logic, helping reduce operational errors and manual control issues by approximately 66–70%.
- Designed backend logic to aggregate multi-table material and inventory data, enabling real-time calculation of operational costs and structured business parameters.
- Implemented transactional logging mechanisms to track user interactions and generate structured audit trails for monitoring, reporting, and system traceability.

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — Stanford University / DeepLearning.AI | [Credential](#)
Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI | [Credential](#)
Tools for Data Science — IBM | [Credential](#)

LANGUAGES

English | Full Professional Proficiency (C1)
Turkish | Native